

Q&A 26

NP vs co-NP, Clique, and Metric TSP Approximation

1 Question 4

Question

Prove that if

$$\mathbf{NP} \neq \mathbf{co-NP},$$

then

$$\mathbf{P} \neq \mathbf{NP}.$$

Proof by Contrapositive

We prove the contrapositive:

$$\mathbf{P} = \mathbf{NP} \implies \mathbf{NP} = \mathbf{co-NP}.$$

Assume $\mathbf{P} = \mathbf{NP}$. Since $\mathbf{P}$ is closed under complement, if $L \in \mathbf{NP}$, then $L \in \mathbf{P}$. Therefore $\bar{L} \in \mathbf{P}$. Since $\mathbf{P} = \mathbf{NP}$, we get $\bar{L} \in \mathbf{NP}$. Hence $L \in \mathbf{co-NP}$.

So

$$\mathbf{NP} \subseteq \mathbf{co-NP}.$$

The same argument applied to complements gives

$$\mathbf{co-NP} \subseteq \mathbf{NP}.$$

Therefore

$$\mathbf{NP} = \mathbf{co-NP}.$$

Thus, if $\mathbf{NP} \neq \mathbf{co-NP}$, then it cannot be true that $\mathbf{P} = \mathbf{NP}$. Hence

$$\mathbf{P} \neq \mathbf{NP}.$$

Write This in Exam

Use contrapositive. If $\mathbf{P} = \mathbf{NP}$, then every language in $\mathbf{NP}$ is in $\mathbf{P}$. Since $\mathbf{P}$ is closed under complement, its complement is also in $\mathbf{P} = \mathbf{NP}$. Hence $\mathbf{NP} = \mathbf{co-NP}$. Therefore, if $\mathbf{NP} \neq \mathbf{co-NP}$, then $\mathbf{P} \neq \mathbf{NP}$.

2 Question 5

Question

Prove that the clique problem is NP-complete.

Problem

$$\mathbf{CLIQUE} = \{\langle G, k \rangle \mid G \text{ has a clique of size at least } k\}.$$

Proof

First, CLIQUE is in **NP**. A certificate is a set S of k vertices. The verifier checks every pair of vertices in S and confirms that the pair is an edge. This is polynomial time.

For NP-hardness, reduce from 3-CNF-SAT. Let

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

be a 3-CNF formula. Construct a graph G as follows:

- for each literal occurrence in each clause, create one vertex;
- connect two vertices by an edge iff they come from different clauses and their literals are not contradictory.

Set

$$k = m.$$

If ϕ is satisfiable, choose one true literal from each clause. The chosen literals are from different clauses and are mutually consistent, so the corresponding m vertices form a clique.

Conversely, if G has a clique of size m , then it contains exactly one vertex from each clause, because vertices from the same clause are not joined. Also, no two selected literals contradict each other, because contradictory literals are not joined. Assign the selected literals to be true. Then every clause is satisfied.

Thus

$$\phi \in \text{3-CNF-SAT} \iff \langle G, m \rangle \in \text{CLIQUE}.$$

The graph has polynomial size. Since 3-CNF-SAT is NP-complete, CLIQUE is NP-hard. Since CLIQUE is also in **NP**, it is NP-complete.

Write This in Exam

Certificate: k vertices. Verify all pairs are adjacent.

Reduction: from 3-CNF-SAT. Make one vertex for each literal occurrence. Join two vertices if they are in different clauses and are not complementary. Set $k = m$, the number of clauses. A satisfying assignment gives one true literal per clause, hence a clique of size m . A clique of size m gives one consistent literal from every clause, hence a satisfying assignment. Therefore CLIQUE is NP-complete.

3 Question 6**Question**

Define approximation ratio. Give an approximation algorithm for the traveling salesperson problem. Prove that under certain constraints the approximation ratio is 2.

Approximation Ratio

For a minimization problem, an algorithm has approximation ratio ρ if for every input,

$$\frac{A}{OPT} \leq \rho,$$

where A is the cost of the solution produced by the algorithm and OPT is the cost of an optimal solution.

Constraint

The 2-approximation guarantee holds for the **metric TSP**, where the graph is complete and the edge weights satisfy the triangle inequality:

$$c(u, w) \leq c(u, v) + c(v, w)$$

for all vertices u, v, w .

MST-Based 2-Approximation Algorithm

1. Compute a minimum spanning tree T of the complete weighted graph.
2. Double every edge of T . This gives an Eulerian multigraph.
3. Find an Euler tour.
4. Shortcut repeated vertices to obtain a Hamiltonian tour.

Approximation Ratio Proof

Let OPT be the cost of an optimal TSP tour. Removing any edge from an optimal tour leaves a spanning tree. Since T is a minimum spanning tree,

$$c(T) \leq OPT.$$

Doubling every edge of T gives an Euler tour of cost

$$2c(T) \leq 2OPT.$$

When we shortcut repeated vertices, the triangle inequality guarantees that the cost does not increase. Therefore the final Hamiltonian tour has cost

$$A \leq 2c(T) \leq 2OPT.$$

Hence

$$\frac{A}{OPT} \leq 2.$$

So the approximation ratio is 2.

Write This in Exam

For metric TSP, compute an MST, double its edges, take an Euler tour, and shortcut repeated vertices. Since

$$c(MST) \leq OPT,$$

the doubled tree has cost at most $2OPT$. Shortcutting does not increase cost by the triangle inequality. Hence the returned tour has cost at most $2OPT$, so the approximation ratio is 2.